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Problem and research gap

Hartmann et al. (2021) have shown that only 5 countries, including Hungary, moved from simple to complex products. The authors themselves leave Hungary out of their case studies, noting that its low wages combined with high complexity and strong role of outsourcing need separate study. This matters because export data can make a country look complex even if the skills stay inside a few industries, and then the catch-up is fragile. To check this, we need to look at skills, not only at products. Earlier relatedness measures such as co-exports or input-output show which industries appear together or trade with each other, but not whether they need similar skills. Neffke et al. (2017) validate labor flows as a skill-relatedness measure. They showed that if many workers switch between two industries, these industries need similar skills. Labor flows connect industries into a network, where industries are nodes and worker switches are links. Balza et al. (2023) took the labor-flow measure to extractive industries in Colombia and added network tools such as centrality and clusters. They found that extractive industries are central in the network and that industries closer to extractives have fewer employees, mostly because they employ fewer women. So the position of an industry in the network matters.

Fleming et al. (2007) show why being connected in the network matters. They found that US regions where inventors were connected through shorter paths and where more inventors belonged to one large connected group had more patents in the following year. This means that knowledge travels through links between people, and parts of the network with few links get little of it. The gap is that we do not know whether Hungary's complex export industries, such as cars and electronics, are connected by labor flows to the rest of the Hungarian economy, or whether they form a separate cluster where the skills stay inside.

Research question

Are complex export industries in Hungary, specifically cars and electronics, connected by labor flows to the rest of the Hungarian economy, or do they form a separate cluster?

Mechanism

Workers learn skills at their job and much of this knowledge is tacit, so it moves with people, as tacit knowledge cannot be written on a paper and is mostly intuitive. When a worker leaves a factory and joins a firm in another industry, the skills move together with this worker. Fleming et al. (2007) show that knowledge spreads inside connected parts of a network, and Balza et al. (2023) show that the position of an industry in the labor-flow network matters. So if complex export industries are well connected, their skills can spread to other industries, which can then also grow and move to more complex products. If they are isolated, the skills stay inside these industries, and if these industries shrink or leave Hungary, the skills may not be used anywhere else in the economy.

Hypothesis

The complex export industries are weakly connected to the rest of the Hungarian economy, because of the strong role of outsourced production. Most workers who leave such industries move to other complex export industries, so these industries form a separate cluster in the labor-flow network, more closed than other manufacturing industries of similar size.

Data and methodology

We propose to use Hungarian linked employer-employee administrative data, such as the Admin4 database, which follows workers and their employers over time. Industries will be the nodes of the network and job-to-job movements of workers between industries will form the links. Following Neffke et al. (2017), the links will be weighted using labor flows adjusted for industry size, so that unusually frequent worker transitions indicate stronger skill relatedness. We will focus on automotive and electronics manufacturing and compare their connections with each other and with the rest of the economy. Community detection will be used to test whether these industries form a separate cluster. Their integration with other industries will be measured by their network strength and by the share of their link weight going outside the cluster, which we will compare with the same share for other manufacturing clusters of similar size.

The main limitation is data access. Admin4 contains the necessary employment histories and employer information but requires restricted research access. As Plan B, we would use Hungarian or EU Labour Force Survey data to study industry transitions at a more aggregated level.

Expected results and impact

We expect automotive and electronics industries to be strongly connected internally but to have relatively weak labor-flow links with other parts of the Hungarian economy. This would support the idea that Hungary's high export complexity does not necessarily imply that the capabilities behind complex production are widely embedded in the domestic economy. If instead these industries have strong links to many other sectors, this would suggest that skills acquired in complex export production diffuse more broadly than expected.

The results would help distinguish between export complexity based mainly on concentrated in a few isolated industries and complexity accompanied by broader domestic capability development. This matters for industrial policy: if important export sectors are isolated, policies encouraging worker mobility, supplier development, training and stronger connections between export industries and the rest of the economy could make the benefits of complex production spread more widely through the Hungarian economy.

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